

August 3, 2026

Dear Editor,

I am pleased to submit my manuscript, "ATLAS: Anytime-valid Testing of Latent Simulators — Sequential Certification of World-Model Faithfulness via E-Process Betting," for consideration as a research article in Pattern Recognition. This manuscript is original, has not been published previously, and is not under consideration elsewhere. I am the sole author and declare no competing interests.

Importance and fit to scope. World models — learned, action-conditioned simulators used for video prediction, autonomous driving, and model-based decision making — are now central to pattern recognition research, yet the field still evaluates them with a single offline number (Frechet Video Distance, reconstruction error, return gap) computed once before deployment. This manuscript shows that such offline scores are structurally blind to how a world model's faithfulness actually behaves: it decays with rollout horizon and drifts over deployment time, especially once the model is embedded in a closed decision loop that endogenously shapes its own test data. We introduce ATLAS, which instead gives a deployed world model a running, anytime-valid statistical certificate — a trust horizon that a downstream planner can use directly to clip its imagination depth. The method is representation-agnostic (pixels, latent codes, or embeddings), requires no exchangeability assumption, and comes with three proved guarantees (uniform validity, an anytime-valid simulation lemma, and an order-optimal detection- delay bound), validated on a controllable synthetic simulator, four real video benchmarks (Moving MNIST, KTH Actions, KITTI driving, and UZH-FPV drone flight), and learned neural world models. This combination of sequential statistical testing, representation learning, and pattern-recognition benchmarks (video, driving, drone imagery) sits squarely within the scope of Pattern Recognition.

Comparison to the state of the art. The manuscript positions ATLAS against, and empirically compares it with, recent representative work, including: 1. Shin, Ramdas, and Rinaldo (2024), "E-detectors: A nonparametric framework for sequential change detection" — the closest anytime-valid change-detection framework, which we specialize and compare against directly (Table 4). 2. Gibbs and Candes (2021), "Adaptive conformal inference under distribution shift," and its 2026 online-conformal descendants (Leong, 2026, "Online shift detection and conformal adaptation for deployed safety classifiers"; Opoku and Banahene, 2026, "Drift-aware spectral conformal prediction for non-exchangeable streaming data") — representative of the conformal-prediction line we compare against as an anytime-valid baseline and discuss as lacking a horizon-resolved, endogeneity-robust guarantee. 3. Assran et al. (2025), "V-JEPA 2: Self-supervised video models enable understanding, prediction and planning" — a state-of-the-art action-conditioned video world model that exemplifies the class of representation ATLAS is designed to certify. 4. NVIDIA et al. (2025), "Cosmos world foundation model platform for physical AI" — a state-of-the-art large-scale physical-AI world-model platform illustrating the deployment setting (offline-evaluated, production world

models) that motivates this work. We are not aware of prior work that gives a horizon-resolved, anytime-valid faithfulness certificate for world models under policy-endogenous data; Section 1.4 (Related work) and Section 5.4 (Comparison with competitor monitors) detail this positioning and the empirical gap quantitatively.

Relationship to prior publications. This manuscript is not an extension of a previously published conference paper; it is original, full-length work.

AI-assistance disclosure. Generative AI (Claude, via Claude Code) was used to help draft, edit, and polish portions of the manuscript text, as disclosed in the manuscript itself. I reviewed all AI-assisted content and take full responsibility for the paper.

Thank you for considering this manuscript. I would be glad to suggest reviewers or provide any further information the editorial process requires.

Sincerely,

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